

# The Impact of Big Data Characteristics on Credit Risk Assessment

Amin Karami<sup>1\*</sup> and Chukwuemeka Igbokwe<sup>1†</sup>

<sup>1\*</sup>Computer Science and Digital Technologies, University of East London (UEL),  
University Way, London, E16 2RD, , UK.

\*Corresponding author(s). E-mail(s): [a.karami@uel.ac.uk](mailto:a.karami@uel.ac.uk);

Contributing authors: [chukwuemekaigbokwe80@gmail.com](mailto:chukwuemekaigbokwe80@gmail.com);

<sup>†</sup>Amin conducted the primary research, analysis, and write-up, while Chukwuemeka contributed to data collection and summarization.

## Abstract

Credit risk assessment is critical for financial institutions' stability and profitability. Traditional methods struggle to assess borrowers with limited credit history or non-traditional income due to insufficient data and outdated models. Big data analytics, characterized by the 5Vs (Volume, Variety, Velocity, Veracity, and Value), offers a transformative solution. This study reviews 50 papers published between 2019 and 2025, demonstrating that leveraging large volumes of diverse data sources (Volume and Variety) with real-time processing (Velocity) and accurate data (Veracity) significantly enhances prediction and financial inclusion (Value). We found that big data reduces biases, refines risk profiles, and provides actionable insights, addressing the limitations of traditional models. The study also analyzes the interconnectedness of Big Data components, including data sources, infrastructure, and the 5Vs, emphasizing the challenges and opportunities. The paper concludes with recommendations for organizations of different sizes on implementing Big Data for credit risk assessment, highlighting a phased approach for adoption.

**Keywords:** Big Data, Credit Risk Assessment, Machine Learning, Financial Inclusion, Data-driven Approaches, Alternative Data, Real-time Processing

## 1 Introduction

Credit risk assessment is pivotal to the stability and profitability of financial institutions, enabling them to evaluate a borrower's likelihood of repayment [1]. Traditional methods, which are primarily dependent on limited data sources such as credit scores and income history, frequently fail, particularly for individuals with negligible credit history [2–4]. These conventional methods are limited by their dependence on fundamental demographic data, which leads to incomplete and occasionally inaccurate evaluations of borrowers' creditworthiness [5, 6]. For instance, traditional credit scores often fail to capture the full financial behavior of borrowers, particularly those who are new to credit markets or have non-traditional income sources. This dependence on limited data points can result in substantial biases, which may prevent otherwise creditworthy individuals from accessing financial services. Moreover, these methods fail to consider the potential for dynamic changes in a borrower's financial situation, such as unexpected expenses or a sudden job loss, which can significantly impact their ability to repay loans [7]. The lack of granularity and timeliness in the data used also means that risk models are often outdated, failing to reflect the current financial environment [8]. Consequently, these limitations pose significant challenges in extending credit to qualified borrowers and managing default risks effectively [9–11].

The characteristics of Volume, Variety, Velocity, Veracity, and Value of Big Data provide a transformative potential to address these limitations. Nevertheless, each component poses unique obstacles. Traditional systems may be overwhelmed by the sheer volume of data, resulting in challenges with storage and processing [7, 12]. The Variety of data, encompassing structured and unstructured sources from social media, utility bills, mobile phone usage, and more, necessitates sophisticated integration techniques to ensure comprehensive analysis [8]. High Velocity requires real-time processing capabilities that many financial institutions still struggle to implement effectively, posing challenges in timely decision-making and risk assessment [9]. It is imperative to guarantee the accuracy of the data, as the inclusion of unreliable or manipulated information can distort risk assessments and undermine decision-making processes [10]. Lastly, deriving Value from Big Data requires advanced analytics and machine learning models to extract actionable insights. This frequently necessitates a substantial investment in technology and expertise, which presents a substantial challenge for numerous institutions [11]. Despite these challenges, leveraging Big Data can lead to more accurate and inclusive credit risk assessments, ultimately enhancing financial stability and extending credit to a broader range of borrowers.

It is impossible to overstate the significance and utility of the 5Vs of Big Data (Volume, Variety, Velocity, Veracity, Value) in credit risk assessment. The volume of data allows financial institutions to analyse enormous quantities of information, thereby revealing patterns and correlations that traditional methods fail to detect [7]. This comprehensive data analysis enables more precise and detailed risk profiling. By integrating non-traditional creditworthiness indicators, such as social media activity, utility payments, and mobile phone usage, the assessment process is enhanced by the diversity of data sources, which provides a more comprehensive understanding of a borrower's financial behaviour [8]. This holistic approach can significantly reduce biases and improve inclusivity in credit lending [3]. The high Velocity at which data is generated and processed enables real-time risk assessment, enabling institutions to promptly respond to fluctuations in a borrower's financial circumstances, thereby reducing the likelihood of defaults [9]. Ensuring the Veracity of data is paramount, as high-quality, accurate data is essential for reliable risk assessments. Advanced data cleansing and validation techniques help maintain data integrity, ensuring that decisions are based on trustworthy information [10]. Finally, the Value derived from Big Data lies in its ability to provide actionable insights through advanced analytics and machine learning models. These insights enable predictive analytics, identifying potential risks before they materialize and allowing for proactive risk management [11]. By leveraging the 5Vs of Big Data, financial institutions can enhance the precision and effectiveness of their credit risk assessments, leading to more stable and inclusive financial systems.

This study systematically reviews 50 papers published between 2019 and 2025 that apply big data analytics to credit risk evaluation. The primary aim of this research is to investigate the impact of big data characteristics on the accuracy and effectiveness of credit risk assessment, providing valuable insights and recommendations to optimize these practices. This research bridges the gap between traditional credit risk assessment methods and the potential offered by big data analytics, thereby contributing to a more inclusive and stable financial system. The paper is structured as follows. Section 2 discusses Big Data's revolutionizing impact on credit risk assessment. Section 3 presents the literature review. Section 4 discusses our findings on the impact of Big Data characteristics on credit risk assessment. Section 5 provides concrete suggestions for the challenges of the 5V's for organizations of different sizes. Finally, conclusion is given in Section 6.

## 2 Big Data's Revolutionizing Impact on Credit Risk Assessment

Big data has revolutionized credit risk assessment by enhancing the methods used to evaluate borrowers' creditworthiness. Traditional credit evaluation methods often rely on limited data sources, which can lead to inaccurate assessments and missed opportunities for both lenders and borrowers [13]. In contrast, big data credit evaluation expands the connotation of credit information, enriching its content and improving decision quality. This shift allows financial institutions to utilize a broader range of data, including non-traditional sources such as shopping history and online behavior patterns, which can significantly enhance credit scoring models [14]. The integration of big data into credit risk analytics enables banks to monitor customer behavior in real-time, facilitating immediate insights and decision-making [15, 16]. This capability is crucial for identifying potential risks and

fraud, as it allows institutions to respond swiftly to changes in customer behavior and market conditions. For instance, systems like CreditEase Financial Cloud leverage big data technologies to provide functions such as anti-fraud measures and real-time loan granting, which are essential for modern credit risk management [17]. Moreover, the development of sophisticated credit scoring models is increasingly reliant on big data techniques. These models utilize a unique combination of datasets, including call-detail records and transaction histories, to create accurate credit scores that reflect the financial credibility of applicants [18]. The ability to analyze vast amounts of unstructured data not only improves the accuracy of credit assessments but also reduces the data threshold for traditional evaluations, making credit more accessible to a wider range of applicants. In addition to enhancing credit risk assessment, big data also plays a vital role in ensuring regulatory compliance. Financial institutions are now better equipped to adhere to laws and regulations, such as those introduced by the Dodd-Frank Act, which affect credit scoring and risk assessment practices [19]. By leveraging big data analytics, banks can ensure that their credit evaluation processes meet regulatory standards while also improving their overall risk management strategies.

## 2.1 Hadoop and Spark Use Cases in Credit Risk Assessment

Hadoop and Spark are two prominent big data technologies that have been effectively utilized in credit risk assessment, offering distinct advantages in processing large datasets and enabling sophisticated analytical models. Both technologies are leveraged to enhance the accuracy and efficiency of credit risk evaluation, which is crucial for financial institutions in managing potential defaults and optimizing lending strategies. The following sections explore specific use cases of Hadoop and Spark in credit risk assessment, highlighting their unique contributions and applications.

### *Apache Spark Use Cases*

- **Fraud Detection:** Apache Spark is employed in credit card fraud detection by integrating machine learning algorithms such as K-Means clustering and C5.0 decision trees. This hybrid approach allows for the classification of normal and fraudulent transactions in large datasets, demonstrating high classification accuracy and suitability for complex data scenarios [20]. Spark's in-memory processing capabilities enable rapid analysis of real-time transaction data, facilitating immediate detection and prevention of fraudulent activities.
- **Real-time Data Processing:** Spark's capability to handle real-time data streams is crucial for near-instantaneous credit risk assessment. This is particularly beneficial in environments where rapid decision-making is required, such as in online lending platforms and high-frequency trading. Spark's streaming functionalities enable continuous ingestion and analysis of data streams, allowing for dynamic credit risk evaluation and immediate response to changing customer behavior or market conditions.
- **Scalability and Performance:** Spark's architecture supports linear scalability, making it ideal for processing large volumes of financial data efficiently. This is evidenced by its application in the ACTUS (Algorithmic Contract Types Unified Standard) approach for financial contract analysis, which shows close to linear scalability in a Spark environment [21]. Spark's ability to distribute data and computations across a cluster enables efficient processing of massive datasets, ensuring timely and accurate credit risk assessments even with growing data volumes.

### *Hadoop Use Cases*

- **Risk Analytics in E-Payments:** Hadoop is utilized in risk analytics within the e-payments industry, where it processes vast amounts of transaction data to identify potential risks. Its MapReduce programming model is particularly effective in handling the volume, variety, and velocity of data typical in this sector [22]. Hadoop's distributed storage and processing capabilities enable efficient analysis of diverse data sources, including transaction logs, customer profiles, and merchant information, to identify patterns and anomalies indicative of fraudulent or high-risk transactions.
- **Credit Scoring and Classification:** In peer-to-peer lending platforms, Hadoop Distributed File System (HDFS) supports the implementation of machine learning models like Logistic Regression for credit scoring. This setup enhances the accuracy and efficiency of credit risk assessments by leveraging diverse data sources, including social media activity, online behavior, and financial history [23]. Hadoop's ability to store and process unstructured data allows for a more comprehensive view of borrower creditworthiness, leading to more informed lending decisions.
- **Comprehensive Risk Management:** Hadoop is often a core component of a hybrid big data architecture used in comprehensive risk management systems for banks. This architecture integrates Hadoop with other processing engines and databases to manage and analyze complex datasets, improving the

performance of risk management systems [24]. Hadoop’s scalability and fault tolerance ensure reliable data storage and processing, while its integration with other technologies enables a holistic approach to risk management, encompassing credit risk, market risk, and operational risk.

### *Performance Benefits*

- **Execution Speed:** Spark is generally faster than Hadoop due to its in-memory processing model. For instance, Spark can execute tasks up to 4.5 times faster than Hadoop in certain machine learning algorithms, such as K-nearest neighbor (KNN) [25]. This speed advantage is crucial for real-time credit risk analytics, where timely data processing is essential for fraud prevention and dynamic credit scoring.
- **Throughput and Speedup:** Spark achieves significant speedup in data processing tasks, with up to 14 times faster execution in TeraSort workloads compared to Hadoop when parameters are optimized. This is particularly beneficial for processing large-scale credit risk data, where high throughput is necessary to handle the volume of transactions and ensure timely risk assessments.
- **Resource Utilization:** Spark’s efficient use of CPU resources and its ability to handle iterative algorithms make it more effective than Hadoop in scenarios requiring repeated data processing, such as credit risk modeling and model training [25]. Spark’s optimized execution engine minimizes data shuffling and redundant computations, leading to improved resource utilization and faster processing times.

### *Scalability Benefits*

- **Linear Scalability:** Both Hadoop and Spark demonstrate strong scalability, but Spark’s architecture allows for almost linear scalability, which is advantageous for large-scale financial risk analytics. This means that as the data volume increases, Spark can scale efficiently without a proportional increase in processing time, ensuring consistent performance even with growing data demands.
- **Data Partitioning and Parallel Processing:** Spark’s ability to partition data and process it in parallel enhances its scalability, allowing it to handle large datasets more effectively. This is crucial for credit risk data, which often involves complex, large-scale datasets with diverse data sources and formats.
- **Fault Tolerance:** Spark’s built-in fault tolerance mechanisms, such as Resilient Distributed Datasets (RDD) lineage, ensure data integrity and recovery, which is vital for maintaining the reliability of credit risk data processing systems. RDD lineage tracks the transformations applied to data, allowing for efficient data recovery in case of node failures, ensuring continuous operation and preventing data loss.

### *Comparative Insights*

While both Hadoop and Spark are integral to credit risk assessment, they serve different purposes and excel in distinct areas. Hadoop is particularly strong in batch processing and storage of large datasets, making it suitable for comprehensive risk management, historical data analysis, and data warehousing. In contrast, Spark’s strength lies in its ability to process data in real-time and support complex machine learning models, which is essential for fraud detection, dynamic credit scoring, and real-time risk monitoring. The choice between Hadoop and Spark often depends on the specific requirements of the financial institution, such as the need for real-time processing versus batch analysis, the complexity of the data being handled, and the desired performance and scalability characteristics.

## **3 Literature Review**

This section represents a 50 diverse perspectives and contributions on Big Data-oriented Credit Risk Assessment, showcasing the remarkable potential of big data to enhance the reliability, efficiency, and inclusivity of credit risk assessment models. From leveraging unconventional data sources like electricity consumption and blockchain-powered credit indices to integrating mobile phone records and social network analytics, these studies demonstrate the multifaceted applications of big data in addressing the limitations of traditional credit risk assessment methods.

Paper [10] explored the use of electricity data for enterprise credit risk assessment as a reliable alternative to traditional financial data. Their study proposed two hypotheses: abnormal fluctuations in electricity consumption indicate production anomalies and higher credit risk, and late payment of electricity bills reflects cash flow issues, suggesting increased credit risk. Empirical testing using logistic regression on a large dataset of electricity consumption and credit defaults for over 8,000 enterprises supported these hypotheses, demonstrating the practical value of using electricity data for predicting credit risk.

Paper [7] discussed integrating big data with blockchain technology to enhance data value and improve predictive analysis for personal credit assessment. The study focused on incorporating moral and regulatory indicators into credit scoring models, aiming to reduce risk in financial applications like P2P loans.

Paper [26] highlighted the importance of the Internet of Things (IoT) in enterprise asset management and credit risk assessment for telecom operators. The study introduced a credit risk assessment model for big data IoT, emphasizing security, stability, and compatibility. The research also showcased the construction of a big data platform and the application of machine learning to improve risk assessment accuracy.

Paper [27] discussed the impact of the Internet credit industry on traditional banking and the importance of developing a risk early warning model using big data technology. The study emphasized the need for a sound financial early warning mechanism to control risks in Internet finance, addressing challenges like supervision and information security.

Paper [8] went into the development of credit risk assessment models for online lending platforms using big data technology. The authors emphasize the importance of data mining techniques in managing credit risk effectively. They utilize various algorithms such as Random Forest (RF), CART, ANN, and C4.5, with RF showing the best performance. The research is conducted using data from the Renrendai platform, processed using R and Python.

Paper [9] presented a novel approach to credit scoring by utilizing mobile phone data and social networks, aiming to enhance financial inclusion through positive credit information. Incorporating mobile phone data significantly improved the statistical performance of credit scoring models, with individual calling behavior being the most predictive of creditworthiness. The study used call-detail records to build call networks and applied social network analytics to propagate influence from prior defaulters, enhancing model performance in terms of AUC and profit.

Paper [28] discussed the limitations of traditional personal credit risk assessment systems and the potential of using Internet social big data to improve these systems. The study established 19 evaluation indicators and used the Analytic Hierarchy Process (AHP) to assign weights, forming a new credit risk assessment index system. The TOPSIS method was applied to simulate data and verify the system's rationality, showing that it can objectively and fairly evaluate credit risk.

Paper [29] discussed the importance of optimizing credit resource allocation in commercial banks using big data technology, which offers advantages in information collection, processing, and decision-making. The study highlighted the challenges faced by commercial banks, such as information asymmetry and the unbalanced allocation of credit resources, which often favor large enterprises over small and micro enterprises. The paper suggested that big data can help reduce information asymmetry, improve credit risk assessment, and enhance the efficiency of credit management by integrating customer data and using advanced data processing techniques.

Paper [5] explored the impact of financial technology, particularly Peer-to-Peer (P2P) lending, on credit scoring and risk assessment in financial institutions. The study highlighted the use of machine learning techniques on big data platforms, specifically within the HDFS environment, to improve credit scoring methods. The research concluded that Logistic Regression outperforms Decision Tree and Random Forest in terms of accuracy, precision, recall, and runtime for credit scoring.

Paper [30] proposed a new credit decision-making system for small and medium-sized enterprises (SMEs) using big data technology and risk assessment to address common credit issues. The system involved a comprehensive evaluation model, cluster analysis, and MATLAB algorithms to minimize credit risk. The research evaluated credit risk based on enterprise strength and reputation, using metrics like rate of return and gross profit margin.

Paper [6] discussed integrating big data with traditional credit reference businesses, highlighting the limitations of traditional credit data processing methods in analyzing massive credit data in the current market. The study presented a prediction model for individual borrowers' credit risk using Logistic Regression, establishing parameters as credit risk evaluation indicators to demonstrate the model's effectiveness. The research concluded that the Logistic Regression model can accurately assess individual borrowers' credit status, aiding financial institutions in avoiding credit risks.

Paper [31] discussed integrating big data and artificial intelligence in the financial industry, focusing on developing a default prediction model for corporate credit risk. The study compared two deep learning models, convolutional neural networks (CNN) and recurrent neural networks (RNN), with traditional machine learning models like random forest and logistic regression. The RNN model was identified as the optimal model, achieving high accuracy, recall, and F1 score.

Paper [2] discussed transforming the financial industry through Internet technology and big data, focusing on developing Internet credit businesses. The study highlighted the challenges faced by China's credit system, such as information asymmetry, which increases credit risk for banks. The paper proposed using big data and neural network algorithms, specifically the BP neural network, to improve credit risk assessment and early warning systems.

Paper [4] focused on the role of peer-to-peer (P2P) lending platforms in modern financial systems and the associated credit risks due to improper supervision. The research introduced a BP neural network-based algorithm for assessing lending risk, comparing its performance with traditional logistic regression methods. The study used large-scale lending data from 2015-2019 to evaluate these algorithms, finding that the BP neural network outperforms logistic regression in reducing investor risk.

Paper [32] discussed the evolution of personal credit evaluation in China, highlighting the shift from traditional methods to big data-driven approaches. This transition aimed to enhance the theoretical framework of credit risk assessment and support China's inclusive financial strategy. The study explored the use of logistic regression for credit risk assessment, emphasizing its advantages like low computational cost and simplicity, while also noting its limitations such as potential underfitting.

Paper [33] introduced a new method for evaluating borrower creditworthiness using Graph Convolutional Neural Networks (GCNNs). The study addressed the limitations of traditional credit risk models, particularly in managing imbalanced datasets and extracting relevant features from complex borrower relationships. The authors proposed a hybrid GCNN model that combines local and global convolutional operations, significantly improving the accuracy of credit risk predictions by effectively capturing the features and relationships of borrowers.

Paper [34] discussed the rapid development of big data technologies and their impact on the financial industry, emphasizing the need for research in credit risk measurement and decision analysis. The study highlighted the challenges and opportunities presented by big data, particularly in transforming and upgrading the financial industry. The paper reviewed various models and theories related to credit risk, including the zeta model, discriminant analysis, and structural models based on capital market theory.

Paper [11] focused on assessing credit risk in online supply chain finance, particularly for commercial banks dealing with small and medium-sized enterprises (SMEs) in the automobile manufacturing industry. The research introduced a credit risk assessment index system that incorporates specific online indicators, enhancing traditional risk evaluation methods. The study employed a nonlinear LS-SVM model for empirical analysis, comparing its performance with the logistic regression model.

Paper [1] reviewed emerging issues related to market power and user privacy in digital commerce, focusing on the implications of personalization by providers of products and services. The study argued that advancements in AI and machine learning enable the effective use of both live and historical behavioral data. The paper integrated the treatment of personalization based on behavioral data with concerns about privacy and market power in digital networks.

Paper [3] focused on the importance of small and micro enterprises in economic and social development, emphasizing the need for scientific credit risk management to support their growth. The research employed machine learning techniques, specifically the XGBoost model, combined with the SMOTE algorithm to handle imbalanced datasets and improve credit risk assessment accuracy. The study identified key factors affecting creditworthiness, such as financial information and operational metrics, and ranked them using the XGBoost classifier.

Paper [35] explored integrating big data mining and artificial intelligence to assess the value of power data assets, focusing on enterprise credit models in the power industry. The study introduced an attentional convolution neural network-based credit rating (ACNNCR) model, utilizing data mining, label system construction, and credit evaluation methods. The research developed power credit label models, including factual, rule-based and predictive labels, based on non-residential user data, establishing 185 features for the ACNNCR model.

Paper [36] focused on improving credit risk assessment models for unlisted technology SMEs, which are crucial for innovation but face challenges with traditional financing models. The study introduced an early warning system by integrating the BP-KMV model with the GARCH (1, 1) model and the Ordinary Least Squares (OLS) method, tailored to the volatility of technology ventures. The model used alternative data sources, such as intellectual property and R&D expenditures, to provide a comprehensive view of an SME's creditworthiness.

Paper [37] emphasized the growing importance of credit risk management in the banking sector due to financial globalization and innovation in the financial industry. The study discussed

constructing a bank credit risk model using optimal indicators and correlation tests. The SMOTE up-sampling method was used for data sampling, and data preprocessing was done with the Min-MaxScaler method. The processed data was input into an improved momentum BP neural network for credit risk assessment, highlighting the interdependence and correlation among features.

Paper [38] addressed the challenge of Enterprise Credit Risk (ECR) assessment, which is crucial for investment decisions and financial regulation. The study proposed a Multi-Structure Cascaded Graph Neural Network (MS-CGNN) framework that enhances enterprise representation learning by utilizing various graph structures, including knowledge graphs and hypergraphs. MS-CGNN incorporated both pairwise and high-order relationships to improve the learning of enterprise representations.

Paper [39] focused on credit risk prediction using machine learning, highlighting its importance for financial institutions to avoid losses from defaults on loans and credit cards. The research addressed the class imbalance problem in datasets used for credit risk prediction, which can lead to biased machine learning models. The study implemented a filter-based feature selection method using Information Gain (IG) to reduce the feature space and improve model performance.

Paper [40] discussed a credit risk prediction method using a hybrid deep learning model combined with sand cat swarm optimization for feature selection, known as SCSOFS-HDL. The SCSOFS-HDL model optimized feature selection from credit risk data and employed a deep LSTM Supervised Autoencoder Neural Network (DLSTM-SANN) for classification. The political optimizer (PO) was used for hyperparameter tuning to enhance the DLSTM-SANN model's performance.

Paper [41] proposed a predictive model for assessing the risk of automotive consumer loans using Generative Adversarial Networks (GANs) and Stacked Autoencoder Neural Networks (SAEs). The study addressed challenges such as high dimensionality, data sparsity, and class imbalance in big data environments. GANs were used to tackle class imbalance, while SAEs were employed for effective feature selection.

Paper [42] discussed the importance of establishing a robust security domain prevention system in the Middle Office, which is crucial for risk management, compliance, and data protection. The study introduced the use of Choquet expectation-based methodologies to optimize security systems by combining multiple criteria and preferences. A Fuzzy Optimization Membership Estimation (FOME) model was proposed for computing feature vectors, using the Flemingo Optimization model for evaluation in the business middle office.

Paper [43] discussed the evolving nature of financial transactions and the increasing volume of data in the banking industry. The study highlighted the necessity for comprehensive risk management strategies to address these challenges. The integration of Artificial Intelligence (AI) and Internet of Things (IoT) data analytics was proposed as a promising approach to enhance risk management in banking operations.

Paper [44] discussed the importance of assessing credit risk in the financial services and lending sector, highlighting its impact on financial stability and societal well-being using Random Forest, KNN, Logistic Regression, SVM, and Decision Tree. The research emphasized that traditional systems like the FICO score have been effective for decades, but the evolving financial landscape demands more innovative and interpretable solutions.

Paper [45] focused on the credit risk assessment (CRA) of Micro, Small, and Medium-sized Enterprises (MSMEs) using a novel model called RGCN-RF, which combines Relational Graph Convolutional Network (RGCN) and Random Forest (RF) algorithms. RGCN was used to identify topological structures and relationships in the data, while RF classified MSMEs based on enterprise embedding vectors generated by RGCN.

Paper [46] focused on assessing credit risks for small and medium-sized enterprises (SMEs) in China using machine learning algorithms. The study utilized performance data and authorized invoice data from 425 SMEs in Chongqing to predict customer default and delinquency. Machine learning methods like logistic regression, random forest, support vector machine, and soft voting ensemble learning were employed to create a prediction classifier.

Paper [47] explored using Machine Learning for credit risk assessment in Multichain Decentralized Finance (DeFi). The study utilized a dataset from multiple blockchains, including Ethereum, and applied data preprocessing techniques to extract DeFi-specific features. Machine learning models like Logistic Regression, Random Forest, XGBoost, CatBoost, LightGBM, and CNN were used to predict wallet liquidations.

Paper [48] addressed the challenge of assessing credit risk in the domestic lending industry, focusing on improving the accuracy of credit assessment models. A multi-stage approach was proposed, involving the K-Means-SMOTE algorithm to address class imbalance, ResNet for feature extraction,

and a two-layer LSTM for learning deep information. The model's performance was further enhanced using the IDWPSO algorithm for optimization.

Paper [49] introduced a financial risk control model called the MSHDS-RS model, which addressed issues in big data risk control technology by using a multi-source heterogeneous data structure and random subspace method. The model enhanced feature fusion optimization through a normalized sparse model, improving the accuracy of financial risk assessments by approximately 3.0 to 3.6 times compared to existing methods.

Paper [50] analyzed credit risk in Internet finance, highlighting its diversification and complexity, which included both financial and non-financial factors. ESG factors, principal component analysis, and the grey correlation method were used to optimize indicators for a comprehensive and accurate system. A Convolutional Neural Network (CNN) model was developed for credit risk warning, with a sub-convolutional network for different data types, improving accuracy and practicality.

Paper [51] addressed the challenge of predicting credit card defaults by integrating deep learning with explainable artificial intelligence (XAI) techniques to enhance model interpretability and practical adoption. The proposed method was evaluated using a real-world dataset and demonstrated competitive prediction accuracy. Key features identified by the model included payment delays and outstanding bill amounts, providing insights into factors driving credit card default risk.

Paper [52] focused on improving credit scoring accuracy by integrating supervised and unsupervised learning methods. An integrated model combining Kohonen's Self-Organizing Maps (SOM) and the Random Forest algorithm was proposed. Key features for model training were identified through feature selection and extraction. The integrated model significantly enhanced the discrimination and prediction accuracy of user credit risk, providing a more reliable tool for financial institutions.

Paper [53] addressed the challenge of predicting credit risk for small and medium-sized enterprises (SMEs) in supply chain finance (SCF) using machine learning techniques. A novel approach combining the Improved Sparrow Search Algorithm (ISSA) and Light Gradient Boosting Machine (LightGBM) was proposed to enhance prediction efficiency and accuracy. ISSA improves upon the original Sparrow Search Algorithm by incorporating fractional calculus concepts and Cauchy-Gaussian mutation to avoid local optima and improve convergence speed.

Paper [54] introduced a credit scoring model using a multitask learning technique based on Siamese neural networks to enhance both predictive power and stability in credit scoring models. The model used personal loan execution data to predict customer defaults while aligning the score distribution with a predefined golden distribution, ensuring stability. The golden distribution was derived from scores generated by a pretrained deep neural network and transformed into a five-grade scale.

Paper [55] discussed the importance of credit risk management in banking, highlighting its impact on profitability and systemic stability. The study emphasized the role of explainable AI (XAI) in credit risk management, which helps in understanding complex data and improving decision-making processes. The article also explored the use of machine learning models, such as tree-enhanced gradient boosting decision trees, for credit scoring and risk assessment.

Paper [56] addressed the challenge of bank credit risk and the importance of identifying qualified credit card holders to enhance bank profitability. The research highlighted the transition from manual labor-intensive screening processes to the use of machine learning (ML) models, which have been improving in accuracy and reliability. The study utilized a dataset of over 40,000 records from a commercial bank to compare dimensionality reduction techniques like PCA and T-SNE, and models such as LightGBM, XGBoost, and Tabnet.

Paper [57] addressed the reluctance of lenders to adopt machine-learning techniques for credit default prediction due to concerns about transparency and explainability. The study introduced credit default prediction models using tree-based ensemble methods, highlighting XGBoost as the most effective model. The research enhanced model explainability by implementing SHapley Additive exPlanations (SHAP) in ML-based credit scoring models, using data from the Lending Club.

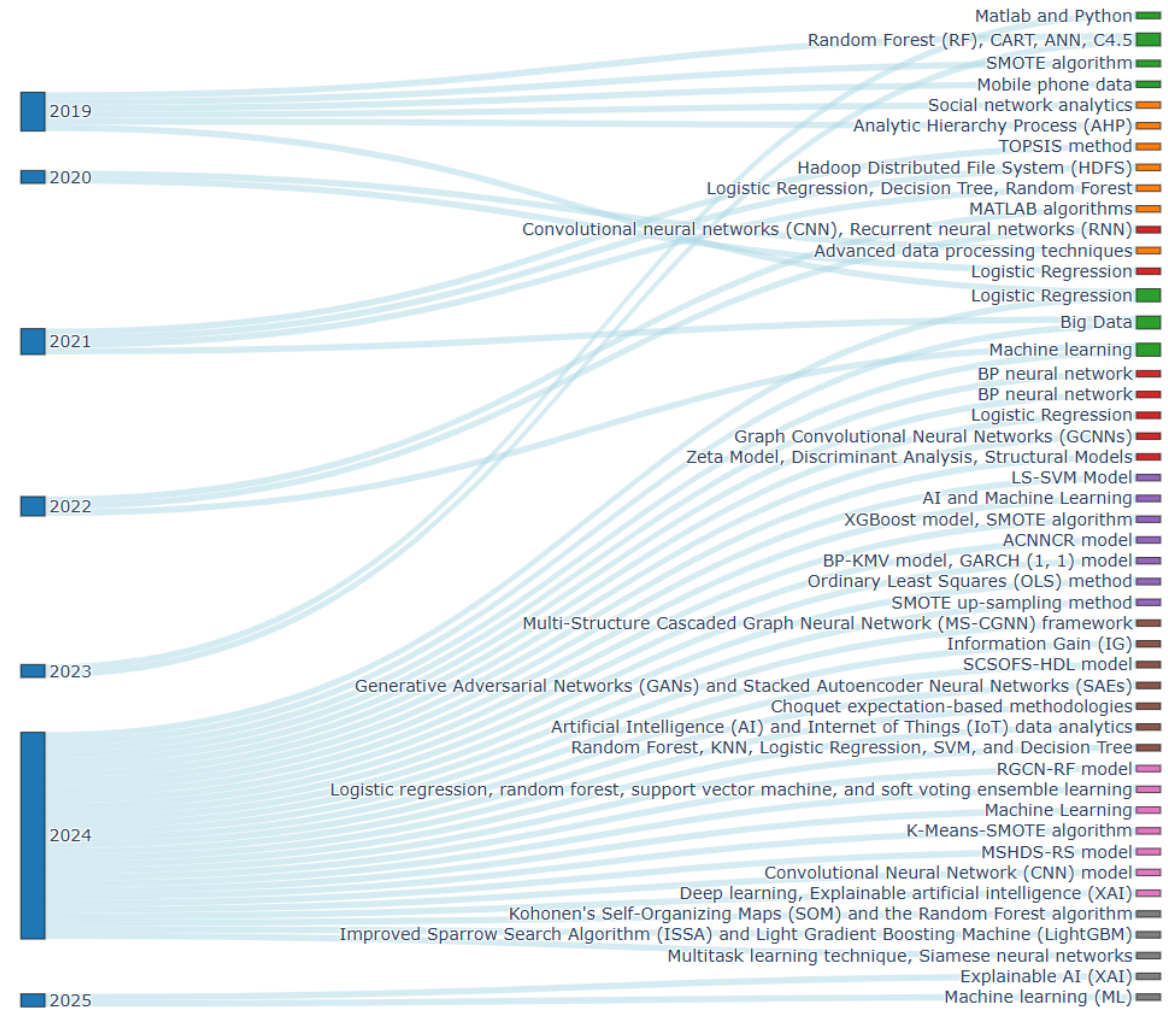
Paper [58] discussed the importance of blockchain technology in secure data storage and credit model sharing, emphasizing its decentralized and untamperable nature. The study introduced EFCS, a federated learning and credit model sharing scheme, which addressed issues like credit data silos and privacy risks. The research proposed a blockchain-based credit data sharing method that ensures data privacy and compliance with data privacy regulations.

Paper [59] focused on enhancing the use of ML models in credit scoring by integrating traditional and non-traditional model risk frameworks. The study addressed the challenge of model interpretability, which is crucial in regulated environments like banking, by proposing a framework that uses

Shapley values to explain ML model predictions. The research was qualitative and evaluated existing model risk frameworks to determine their suitability for validating ML models in credit scoring.

### 3.1 A summary of Findings: From Traditional Statistics to Advanced AI

Figure 1 illustrates our observations and findings on the evolution of financial risk assessment techniques from 2019 to 2025. The nodes (boxes) represent either years or techniques, with their size reflecting the number of connections they have - larger nodes indicate techniques that were either widely adopted or frequently referenced in the literature. This diagram reveals a clear progression from basic statistical methods to more sophisticated approaches. In the early years (2019-2020), the focus was primarily on fundamental statistical methods and traditional machine learning, but the field rapidly evolved through 2021-2023 with increased adoption of deep learning and neural networks. The most significant technological advancement is evident in 2024-2025, marked by the emergence of advanced AI techniques, explainable AI, and hybrid models.



**Fig. 1** Credit Risk Assessment Methodologies and Approaches Over the Time

From analyzing the Figure 1, Big Data's role in credit risk assessment has evolved significantly but faces notable challenges. Initially appearing around 2019-2020, Big Data analytics intersected with mobile phone data and social network analytics, marking the beginning of alternative data sources in credit assessment. The integration of HDFS and advanced data processing techniques in subsequent years highlights the infrastructure needed to handle massive datasets. However, the diagram reveals several ongoing challenges: the need to effectively process and analyze unstructured data sources, the complexity of integrating traditional credit assessment methods with Big Data approaches, and the requirement for scalable solutions that can handle real-time data processing.

The emerging trends show a shift toward combining Big Data with AI techniques, particularly in areas like federated learning and credit model sharing, suggesting a future where distributed data analysis becomes crucial. Notable gaps in the diagram indicate potential areas for development, such as standardization of Big Data analytics processes in credit risk assessment and the need for more robust data privacy frameworks when handling large-scale financial data. The relatively limited direct connections between Big Data nodes and newer AI techniques in the diagram also suggests an opportunity for further integration of Big Data analytics with emerging technologies like explainable AI and multitask learning techniques.

## 4 The Impact of Big Data Characteristics on Credit Risk Assessment

### 4.1 Data-Driven Approaches

The reviewed literature highlights a diverse range of data-driven approaches used in credit risk assessment:

- Machine Learning (ML) Models: ML models are extensively used for their ability to learn complex patterns and relationships in data. The commonly used models include:
  1. Logistic Regression for its simplicity and interpretability.
  2. Support Vector Machines (SVM) for their effectiveness in high-dimensional spaces.
  3. Neural Networks (ANN, CNN, RNN) and Deep Learning for their ability to capture complex patterns in large datasets.
- Statistical Models: Traditional statistical models also play a role in credit risk assessment:
  1. Discriminant analysis for classifying borrowers into different risk categories.
  2. Structural models based on capital market theory for estimating the probability of default based on a borrower's financial structure.
  3. The Zeta model for predicting credit risk based on historical default rates.
- Hybrid Models: Combining different models can leverage their strengths and improve overall performance.
  1. Integrating BP-KMV with GARCH (1, 1) and OLS to capture the volatility of technology ventures.
  2. Combining RGCN with Random Forest to identify topological structures and relationships in data for classifying MSMEs.
- Optimization Algorithms: These algorithms are used to enhance the performance and efficiency of credit risk models.
  1. SMOTE for addressing class imbalance in datasets.
  2. SCSOFS-HDL for optimizing feature selection.
  3. IDWPSO for fine-tuning model parameters.
- Graph-Based Approaches: These approaches are gaining traction for their ability to analyze complex relationships in data.
  1. Graph Convolutional Neural Networks (GCNNs) for capturing the features and relationships of borrowers.
  2. Multi-Structure Cascaded Graph Neural Networks (MS-CGNN) for enhancing enterprise representation learning by utilizing various graph structures.
- Large Language Models (LLMs): LLMs are increasingly being utilized in credit risk assessment for their ability to process and extract insights from unstructured financial data.
  1. LLMs can analyze regulatory filings, financial news, and earnings reports to detect potential risks and emerging financial trends.
  2. They enhance automated credit scoring systems by incorporating qualitative factors from textual data, improving the robustness of risk predictions.
  3. Fine-tuned domain-specific LLMs can assist in assessing borrower credibility by summarizing historical financial behaviors and legal documents.
- Explainable AI (XAI): As credit risk models become more complex, the need for transparency and interpretability increases.
  1. SHAP (SHapley Additive exPlanations) is used to explain the contribution of features to specific outputs, enhancing the understanding of model predictions.

### 4.2 Data Sources and Integration

Effective credit risk assessment relies on diverse data sources and their seamless integration:

### ***Traditional Financial Data:***

This remains a cornerstone of credit risk assessment, providing valuable insights into a borrower's financial history and stability. Key sources include

- Financial statements, such as balance sheets, income statements, and cash flow statements, to assess a borrower's financial health.
- Credit reports, which provide a summary of a borrower's credit history, including outstanding debts, payment history, and credit utilization.
- Loan applications, which contain detailed information about the borrower's purpose for the loan, employment history, and income.
- Payment histories, which track a borrower's past payment behavior on loans and credit cards, providing insights into their creditworthiness.

### ***Alternative Data:***

With the rise of Big Data, non-traditional data sources are increasingly used to complement traditional financial data. These sources include:

- Social media data, which can reveal insights into a borrower's lifestyle, social connections, and spending habits.
- Mobile phone usage data, which can provide information about a borrower's location, communication patterns, and app usage.
- Electricity consumption data, which can be used to assess the financial health of businesses.

### ***Big Data Platforms:***

These platforms provide the infrastructure for storing, processing, and analyzing large volumes of data from various sources.

- Hadoop and Spark are widely used for their distributed computing capabilities, enabling efficient handling of massive datasets.

### ***Data Preprocessing:***

This is a crucial step in ensuring data quality and preparing it for analysis. Key techniques include:

- Standardization, which transforms data into a common format for comparison and analysis.
- Balancing, which addresses class imbalance in datasets to prevent biased model training.
- Normalization, which scales data to a specific range to improve model performance.
- Feature selection, which identifies the most relevant features for credit risk assessment, reducing dimensionality and improving model efficiency.

## **4.3 Technological Infrastructure**

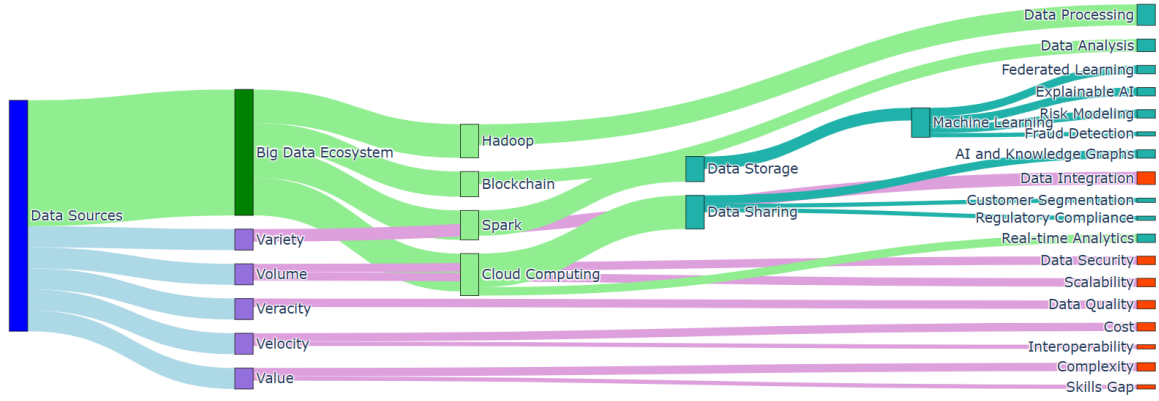
The technological infrastructure plays a vital role in enabling Big Data analytics for credit risk assessment:

- Cloud Computing: Cloud-based solutions offer scalability, flexibility, and cost-effectiveness for Big Data analytics. Cloud platforms like AWS, Azure, and GCP provide on-demand access to computing resources, storage, and analytics tools.
- Distributed Computing: This is essential for processing massive datasets efficiently. Frameworks like Hadoop and Spark distribute the computational workload across multiple nodes, enabling parallel processing and faster analysis.
- Real-time Analytics: Timely risk assessment is crucial in today's dynamic financial landscape. Real-time analytics technologies enable continuous data ingestion, processing, and analysis, providing up-to-date insights for decision-making.

## **4.4 Application Areas**

Big Data analytics is transforming credit risk assessment across various application areas:

- Consumer Lending: Assessing credit risk for individuals applying for loans or credit cards is a major application area. Big Data analytics helps lenders evaluate creditworthiness more accurately, personalize loan offers, and prevent fraud.



**Fig. 2** Big Data Sources on Credit Risk Assessment

- **SME Lending:** Small and medium-sized enterprises (SMEs) often face challenges in accessing credit due to limited financial history. Big Data analytics can help lenders assess the credit risk of SMEs more comprehensively by considering alternative data sources and non-traditional factors.
- **Corporate Finance:** Predicting default risk for large corporations is crucial for investors and lenders. Big Data analytics can provide a more granular view of corporate financial health, enabling better risk assessment and investment decisions.
- **Supply Chain Finance:** Assessing credit risk within supply chain networks is essential for ensuring smooth operations and mitigating disruptions. Big Data analytics can help identify potential risks and vulnerabilities in the supply chain, enabling proactive risk management.
- **Decentralized Finance (DeFi):** The emergence of DeFi poses new challenges and opportunities for credit risk assessment.

## 4.5 Emerging Trends

The field of credit risk assessment is constantly evolving, with several emerging trends shaping its future:

- **AI and Knowledge Graphs:** Integrating AI with knowledge graphs can enhance financial risk management by providing a more comprehensive view of borrower relationships and dependencies.
- **Blockchain Technology:** Blockchain can revolutionize credit risk assessment by enabling secure data storage and sharing among different stakeholders. Its decentralized and tamper-proof nature ensures data integrity and reduces fraud risk.
- **Federated Learning:** This approach allows for collaborative model training without sharing sensitive data, addressing privacy concerns and promoting data security.
- **Explainable AI:** As credit risk models become more complex, the need for transparency and interpretability increases. Explainable AI (XAI) techniques can help demystify model predictions, build trust, and ensure compliance with regulations.
- **Large Language Models (LLMs):** The use of LLMs in credit risk assessment can enhance decision-making by extracting insights from vast amounts of unstructured data, such as financial reports, news, and legal documents. LLMs can assist in automating risk analysis, identifying emerging risks, and improving the efficiency of credit evaluation processes.

## 4.6 Big Data 5V's

Figure 2 provides a comprehensive and impactful visualization of the key findings from the literature review regarding Big Data in credit risk assessment. It highlights the interconnectedness of various components, including data sources, Big Data infrastructure, the 5Vs of Big Data, challenges, and future directions. The diagram effectively illustrates the flow of information from data sources, through various data processing and analytical techniques, to different application areas. This flow underscores the importance of integrating diverse data sources and leveraging appropriate technologies to extract valuable insights for credit risk assessment. The diagram emphasizes the crucial role of Big Data platforms and infrastructure in enabling efficient data processing and analysis. It also

**Table 1** A summary of the findings categorized by the 5Vs

| Category | Summary                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Volume   | Many studies involved large datasets from diverse sources, including electricity consumption records, financial transactions, social media data, and mobile phone records. Some studies explicitly mentioned the need for advanced data processing techniques to handle the large volumes of data. This highlights the increasing importance of volume in Big Data for credit risk assessment.                                                                              |
| Variety  | Studies utilized various data types, including structured (e.g., financial transactions), semi-structured (e.g., social media data), and unstructured data (e.g., text, images). The integration of diverse data sources, such as social media, mobile phone data, and IoT sensor data, was a common theme, emphasizing the growing importance of variety in capturing a holistic view of creditworthiness. This also presents challenges in data integration and analysis. |
| Velocity | While many studies implicitly dealt with high-velocity data, such as real-time financial transactions and social media feeds, explicit discussions on velocity were limited. Some studies mentioned the need for real-time or near real-time data processing capabilities, especially in the context of high-frequency trading and fraud detection. This suggests a growing need for advanced techniques to handle the speed of data generation and processing.             |
| Veracity | Several studies emphasized the importance of data quality and reliability for accurate credit risk assessment. Techniques like data cleaning, preprocessing, and validation were often mentioned to ensure data accuracy and consistency. The increasing use of alternative data sources also brings challenges in assessing data trustworthiness and reliability.                                                                                                          |
| Value    | The studies demonstrated the value of Big Data in improving credit risk assessment through enhanced prediction accuracy, better risk management, and more informed decision-making. The use of Big Data analytics can lead to more efficient credit scoring, better identification of risky borrowers, and improved fraud detection. This highlights the potential of Big Data to transform credit risk assessment and contribute to financial stability.                   |

highlights the significance of the 5Vs of Big Data in understanding the challenges and opportunities associated with leveraging Big Data for credit risk assessment.

The diagram incorporates nodes representing the challenges and future directions identified in the literature review. This inclusion provides a holistic view of the Big Data landscape in credit risk assessment, acknowledging the limitations and potential areas for improvement. The interconnectedness of various components in the diagram underscores the complex and dynamic nature of Big Data in credit risk assessment. It highlights the need for a comprehensive approach that considers not only the technical aspects but also the challenges and opportunities associated with data management, analysis, and interpretation.

Table 1 briefly summarizes the impact of big data's 5Vs on credit risk assessment. It highlights the growing importance of each characteristic, the challenges they present, and the value they bring to the field. The table emphasizes the need for advanced techniques to handle the scale and complexity of big data, while underscoring its potential to revolutionize credit risk assessment and contribute to a more stable financial system. Table 2 summarizes the extent to which the 50 reviewed papers incorporate the 5V's and the usage of Spark/Hadoop frameworks. A dash (X) indicates "zero contribution," a circle (●) indicates "partial contribution," and a checkmark (✓) means "full consideration." This table provides a concise overview of how the 5Vs of Big Data are applied in credit risk assessment across various research studies. A striking observation is the predominant focus on Variety, with a majority of studies incorporating diverse data sources to enhance credit risk models. This highlights a shift towards more holistic assessments that go beyond traditional financial data. However, the table also reveals limited emphasis on Velocity. Real-time data processing, crucial for dynamic risk assessment and fraud prevention, appears to be under-explored. This suggests a potential gap in research and practice, especially given the increasing availability of real-time data streams. Furthermore, while most studies acknowledge the importance of Veracity, ensuring data quality and reliability, the table indicates a varying degree of attention to this aspect. This underscores the need for more rigorous data quality management frameworks in credit risk assessment. Finally, the table demonstrates the significant Value derived from applying Big Data in credit risk assessment. Studies consistently report improved prediction accuracy, enhanced risk management, and better-informed decision-making. This reinforces the transformative potential of Big Data in the financial industry.

#### 4.7 Interconnected Nature of Big Data in Credit Risk Assessment

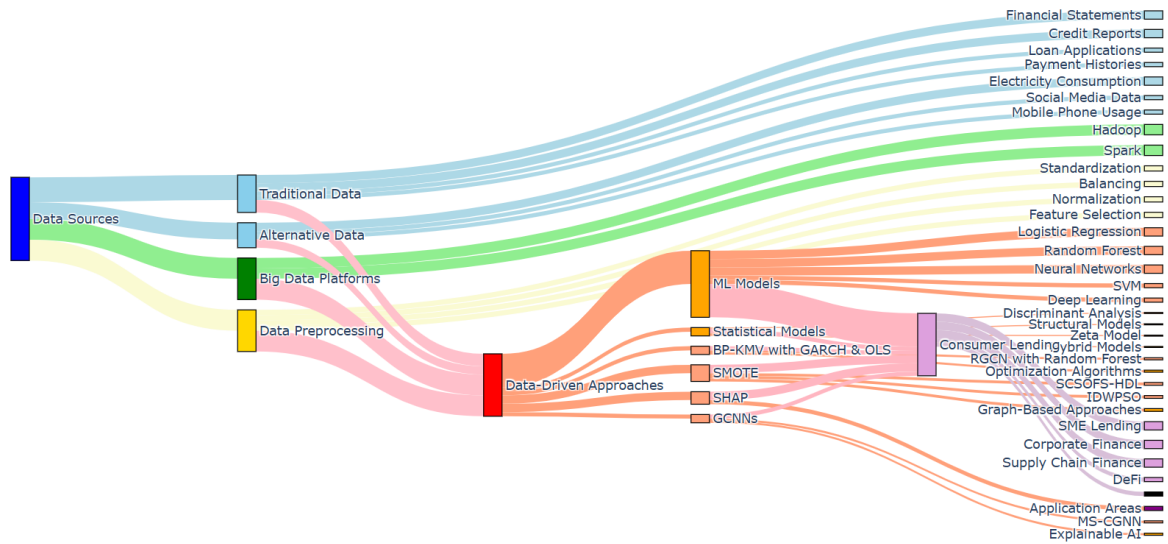
Figure 3 visually represents the flow of information and relationships between different components of Big Data in credit risk assessment.

**Table 2** The Application of the 5 Vs of Big Data in Reviewed Studies

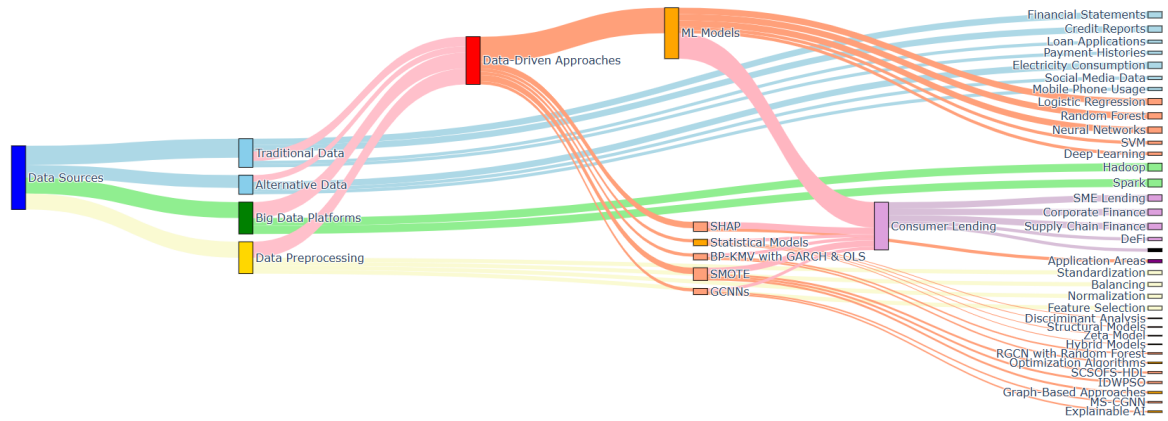
| Paper      | Volume | Variety | Velocity | Veracity | Value | Big Data Ecosystem |
|------------|--------|---------|----------|----------|-------|--------------------|
| [9], 2019  | X      | X       | X        | ●        | ●     | X                  |
| [28], 2019 | X      | ●       | X        | X        | ●     | X                  |
| [5], 2019  | ✓      | ✓       | X        | ✓        | ✓     | ✓                  |
| [11], 2020 | X      | X       | X        | X        | X     | X                  |
| [4], 2020  | ●      | ✓       | X        | ✓        | ✓     | X                  |
| [29], 2021 | X      | ●       | X        | ●        | ●     | X                  |
| [30], 2021 | ●      | ●       | X        | ✓        | ✓     | X                  |
| [2], 2021  | ●      | ●       | ●        | ●        | ●     | X                  |
| [32], 2021 | ●      | ✓       | X        | ✓        | ✓     | X                  |
| [60], 2021 | ●      | ✓       | X        | X        | X     | X                  |
| [7], 2022  | X      | X       | X        | ●        | ●     | X                  |
| [26], 2022 | X      | ✓       | X        | ●        | ●     | X                  |
| [27], 2022 | X      | X       | X        | ●        | ●     | X                  |
| [34], 2022 | X      | ✓       | X        | X        | X     | ●                  |
| [6], 2023  | X      | X       | X        | X        | ●     | X                  |
| [31], 2023 | ●      | ✓       | X        | ✓        | ✓     | ✓                  |
| [10], 2023 | ●      | ✓       | X        | ✓        | ✓     | X                  |
| [61], 2023 | ●      | ✓       | X        | X        | ●     | X                  |
| [57], 2024 | X      | X       | X        | X        | X     | X                  |
| [1], 2024  | X      | X       | X        | X        | X     | X                  |
| [55], 2024 | ●      | ✓       | X        | X        | ✓     | X                  |
| [8], 2024  | X      | ●       | X        | ●        | ●     | X                  |
| [59], 2024 | X      | X       | X        | X        | X     | X                  |
| [39], 2024 | ●      | ✓       | X        | ✓        | ✓     | X                  |
| [3], 2024  | ●      | ✓       | X        | ✓        | ✓     | X                  |
| [44], 2024 | X      | X       | X        | X        | X     | X                  |
| [46], 2024 | ✓      | ●       | X        | X        | X     | X                  |
| [49], 2024 | X      | ●       | X        | ●        | X     | X                  |
| [36], 2024 | ●      | ✓       | ●        | ●        | ●     | X                  |
| [45], 2024 | ●      | ✓       | X        | ✓        | ✓     | X                  |
| [47], 2024 | ●      | ✓       | ●        | ✓        | X     | ●                  |
| [50], 2024 | X      | ✓       | X        | X        | ✓     | X                  |
| [40], 2024 | X      | ●       | X        | X        | X     | X                  |
| [38], 2024 | ✓      | ●       | X        | ●        | X     | X                  |
| [33], 2024 | ●      | ✓       | ●        | ✓        | ✓     | X                  |
| [51], 2024 | X      | X       | X        | X        | X     | X                  |
| [43], 2024 | X      | X       | X        | X        | X     | X                  |
| [48], 2024 | ●      | X       | X        | ✓        | ✓     | X                  |
| [35], 2024 | X      | ✓       | ●        | ✓        | ✓     | X                  |
| [52], 2024 | X      | X       | X        | X        | X     | X                  |
| [37], 2024 | X      | X       | ●        | ✓        | ✓     | X                  |
| [58], 2024 | X      | ●       | X        | ✓        | ✓     | ●                  |
| [42], 2024 | X      | X       | X        | X        | X     | X                  |
| [56], 2024 | ✓      | X       | X        | ✓        | ✓     | X                  |
| [62], 2024 | ●      | ●       | X        | ●        | X     | X                  |
| [41], 2025 | X      | ●       | X        | ✓        | ●     | X                  |
| [53], 2025 | X      | X       | X        | X        | X     | X                  |
| [54], 2025 | X      | X       | X        | X        | X     | X                  |
| [63], 2025 | ●      | ●       | X        | X        | X     | X                  |
| [64], 2025 | X      | X       | X        | X        | ✓     | X                  |

- Nodes: Each node in the diagram represents a specific element, such as a data source, data processing technique, analytical approach, or application area. The size of a node is influenced by its relative importance or frequency in the reviewed literature, with larger nodes indicating greater prominence.
- Links: The links connecting the nodes represent the flow of information or relationships between the elements. The thickness of a link corresponds to the weight assigned to that connection, indicating its relative strength or importance. Thicker links represent stronger or more frequent connections.
- Weights: The weights assigned to the links are based on a critical analysis of the literature, considering the frequency of mention, the emphasis placed on specific connections, and the overall impact on credit risk assessment. Higher weights indicate greater importance or influence.

The diagram highlights the interconnected nature of Big Data in credit risk assessment. It shows how different data sources are integrated, processed, and analyzed using various techniques to address specific application areas. The diagram also emphasizes the importance of emerging trends, such as AI, blockchain, and explainable AI, in shaping the future of credit risk assessment. By visualizing these complex relationships, the diagram provides a comprehensive and impactful overview of the key



**Fig. 3** Big Data in Credit Risk Assessment: Critical Analysis



**Fig. 4** The Key Elements and Their Interrelationships Within the Big Data Credit Risk Assessment Ecosystem

findings from the literature review. It helps to identify the most important elements and connections, understand the flow of information, and gain insights into the evolving landscape of credit risk assessment in the Big Data era.

## 4.8 Additional Observations

Figure 4 provides a visual representation of the key elements and their interrelationships within the big data credit risk assessment ecosystem. It effectively illustrates the flow of information and methodologies, starting from diverse data sources and culminating in various analytical approaches and their practical applications.

1. **Machine Learning Dominance:** The diagram unequivocally demonstrates the prevalent use of ML models in credit risk assessment. The significant weight assigned to the link connecting "Data-Driven Approaches" to "ML Models" underscores their widespread adoption and importance in the field. Furthermore, within ML models, algorithms like Logistic Regression and Random Forest exhibit greater prominence, as evidenced by their comparatively larger link weights, suggesting their frequent utilization. This observation aligns with the increasing trend of leveraging ML's predictive power for more accurate credit risk assessments.
2. **Alternative Data Ascendancy:** The diagram highlights the rising integration of alternative data sources, such as social media activity and mobile phone usage patterns, into credit risk assessment processes. While traditional data sources like financial statements and credit reports

remain crucial, the substantial flow towards alternative data signifies its growing value in providing a more holistic and nuanced understanding of borrowers' creditworthiness. This trend reflects the recognition that alternative data can capture unique insights not available in traditional financial data, leading to more comprehensive risk profiles.

3. **Data Preprocessing Imperative:** The diagram emphasizes the indispensable role of data preprocessing in preparing data for analysis. The strong connection between "Data Sources" and "Data Preprocessing" underscores the significance of tasks like standardization, balancing, normalization, and feature selection in ensuring data quality and suitability for modeling. This observation reinforces the notion that effective data preprocessing is a cornerstone of accurate and reliable credit risk assessment, as it addresses issues like data inconsistency, noise, and redundancy, which can hinder model performance.
4. **Application Diversity:** The diagram showcases the wide-ranging applications of big data credit risk assessment across various financial domains, including consumer lending, SME lending, corporate finance, supply chain finance, and decentralized finance (DeFi). This diversity underscores the adaptability and versatility of these techniques in addressing different credit risk challenges across various lending segments and financial products.
5. **Emerging Trends:** The diagram captures nascent trends in the field, such as the increasing interest in explainable AI (XAI) for enhanced transparency and interpretability of credit risk models, and the exploration of graph-based approaches like Graph Convolutional Neural Networks (GCNNs) for capturing complex relationships within financial data. Additionally, Large Language Models (LLMs) are gaining traction for their ability to process vast amounts of unstructured financial data, such as regulatory filings, news reports, and legal documents, aiding in risk analysis and decision-making. While these areas currently have smaller link weights compared to more established methods, their inclusion signifies their growing relevance and potential for future development in credit risk assessment.

## 5 Recommendations for Organizations on Implementing Big Data for Credit Risk Assessment

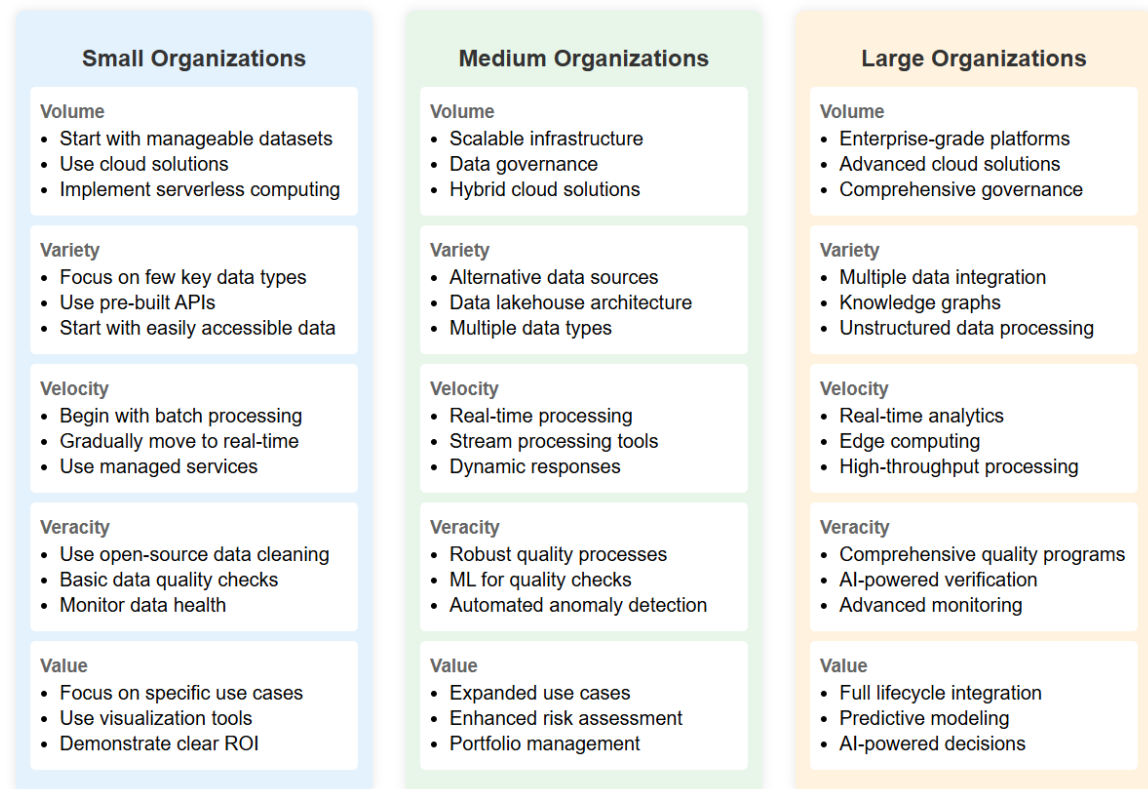
Organizations of all sizes can leverage Big Data to enhance their credit risk assessment processes. However, the implementation journey differs significantly based on an organization's resources and technical capabilities. This section provides tailored recommendations, categorized by the 5Vs of Big Data, while addressing the specific challenges faced by smaller institutions and the complexities of handling diverse, real-time data at scale. The overview of the proposed recommendations is depicted in Figure 5. The guiding principle behind these recommendations is a phased approach to implementation. Organizations should start with manageable solutions, gradually increasing complexity as their capabilities and needs evolve. This allows for gaining experience, building confidence, and demonstrating value at each stage before progressing to more advanced solutions. This phased approach is particularly crucial for smaller institutions, enabling them to adopt big data technologies incrementally while minimizing risks and maximizing returns on investment.

### 5.1 Small Organizations

Small financial institutions often operate with limited budgets and technical expertise. Successfully adopting big data technologies requires careful consideration of cost-effective solutions and strategies that maximize existing capabilities.

#### 5.1.1 Volume

- **Start Small, Scale Gradually:** Begin by focusing on manageable datasets and gradually increase data volume as your organization's capabilities and infrastructure mature. Cloud-based solutions like AWS, Azure, and GCP offer cost-effective storage and processing options, allowing for scalability without significant upfront investment. To further minimize costs, consider utilizing free tiers or pay-as-you-go models offered by cloud providers.
- **Embrace Serverless Computing:** Serverless computing platforms like AWS Lambda and Azure Functions provide an excellent entry point for small institutions. They handle data processing tasks without the need to manage servers, offering scalability and cost-efficiency by charging only for the computing resources consumed. This eliminates the need for dedicated IT staff and reduces infrastructure costs.



**Fig. 5** An Overview of Recommendations for Organizations on Implementing Big Data for Credit Risk Assessment

### 5.1.2 Variety

- **Prioritize Key Data Types:** Initially, focus on integrating a few key data types beyond traditional financial data. Prioritize those with the highest potential impact, such as social media data or mobile phone usage patterns, which can provide valuable insights into borrower behavior. Leverage pre-built APIs from data providers like Plaid to simplify the integration of bank transaction data and reduce development time.
- **Address Integration Challenges:** Integrating diverse data sources can be technically challenging, especially for smaller institutions with limited IT resources. Start with data sources that are readily accessible and require minimal preprocessing. Consider using data virtualization tools to create a unified view of data without the need for physical data integration.

### 5.1.3 Velocity

- **Start with Batch Processing:** Initially, focus on batch processing of data at regular intervals to gain insights and build analytical capabilities. This approach is less demanding in terms of infrastructure and technical expertise compared to real-time processing.
- **Gradual Transition to Real-time:** As real-time needs arise, explore stream processing tools like Apache Kafka or Amazon Kinesis, which offer managed services to simplify deployment and management. However, be mindful of the infrastructure costs and technical expertise required for real-time processing.

### 5.1.4 Veracity

- **Ensure Data Quality:** Prioritize data cleaning and validation to ensure data accuracy and reliability. Utilize open-source tools like Great Expectations or Apache Spark for data quality checks, which can be implemented with minimal cost.
- **Data Quality Monitoring:** Consider data observability platforms like Monte Carlo or Acceldata to proactively monitor data health and identify potential issues. While these platforms may involve costs, they can prevent costly errors and improve the reliability of credit risk assessments.

### 5.1.5 Value

- **Focus on Specific Use Cases:** Start with focused use cases that demonstrate tangible value, such as identifying high-risk applicants or automating credit scoring processes. This allows for a clear demonstration of the benefits of big data analytics and justifies further investment.
- **Effective Communication:** Employ visualization tools like Tableau or Power BI to effectively communicate insights derived from big data analytics to stakeholders and demonstrate the value of these initiatives.

## 5.2 Medium Organizations

As organizations grow, they can leverage their increased resources and expertise to implement more sophisticated big data solutions for credit risk assessment.

### 5.2.1 Volume

- **Invest in Scalable Infrastructure:** Invest in scalable infrastructure, either on-premise or in the cloud, to accommodate growing data volumes and ensure efficient processing. Implement data governance frameworks using tools like Collibra or Alation to effectively manage data assets and maintain data quality.

### 5.2.2 Variety

- **Expand Data Sources:** Explore a wider range of alternative data sources, including social media, mobile phone usage, IoT sensor data, and public records. Consider adopting a data lakehouse architecture, which combines the flexibility of data lakes with the structure of data warehouses, to efficiently store and analyze diverse data types.

### 5.2.3 Velocity

- **Incorporate Real-time Processing:** Incorporate real-time processing capabilities to enable faster credit decisions, enhance fraud detection, and respond dynamically to changing market conditions. Explore Apache Flink for stateful stream processing and complex event processing (CEP) to analyze high-velocity data streams and identify patterns in real-time.

### 5.2.4 Veracity

- **Implement Robust Data Quality Processes:** Implement robust data quality processes, including data profiling, cleansing, and validation, to ensure data accuracy and consistency. Leverage machine learning techniques for automated data quality checks and anomaly detection to identify potential data issues proactively.

### 5.2.5 Value

- **Expand Use Cases:** Expand the use of big data analytics to encompass a wider range of credit risk management functions, such as customer segmentation, risk pricing, and portfolio management. This will enable more sophisticated risk assessments and contribute to improved business outcomes.

## 5.3 Large Organizations

Large organizations have the resources and infrastructure to implement comprehensive big data solutions for credit risk assessment, incorporating cutting-edge technologies and advanced analytics.

### 5.3.1 Volume

- **Leverage Enterprise-grade Platforms:** Leverage enterprise-grade big data platforms like Hadoop and Spark, along with cloud-based solutions like Databricks and Snowflake, to handle massive data volumes and complex analytical workloads. Implement robust data governance and security protocols to ensure compliance with regulations like GDPR and CCPA.

### 5.3.2 Variety

- **Integrate Diverse Data Sources:** Integrate a wide variety of data sources, including real-time data feeds, external data marketplaces, and unstructured data like text and images. Utilize knowledge graphs, such as Neo4j or Amazon Neptune, to connect and analyze relationships within data, providing a more comprehensive view of borrowers and their networks.

- **Address Scalability Challenges:** Integrating and processing diverse data at scale presents significant technical challenges. Implement robust data pipelines with distributed processing frameworks like Spark to handle large data volumes and ensure efficient data flow. Consider using data virtualization techniques to create a unified view of data without the need for physical data integration, which can be complex and resource-intensive.

### 5.3.3 Velocity

- **Implement Real-time Analytics:** Implement real-time analytics capabilities for immediate credit decisions, proactive fraud prevention, and dynamic risk pricing. Explore edge computing to process data closer to the source, reducing latency and enabling faster insights.
- **Handle Real-time Data Challenges:** Processing real-time data at scale requires specialized infrastructure and expertise. Consider using stream processing platforms like Apache Kafka or Amazon Kinesis, which offer high throughput and fault tolerance. Implement real-time data quality checks to ensure data accuracy and prevent erroneous decisions based on flawed data.

### 5.3.4 Veracity

- **Establish a Comprehensive Data Quality Program:** Establish a comprehensive data quality program that encompasses data lineage tracking, data quality monitoring, and automated data quality checks. Leverage machine learning and AI for advanced data quality assessment, anomaly detection, and explainability to ensure data accuracy and transparency.

### 5.3.5 Value

- **Utilize Big Data Across the Credit Risk Lifecycle:** Utilize big data analytics across the entire credit risk lifecycle, including stress testing, scenario modeling, and regulatory reporting. Drive innovation by employing advanced analytics, predictive modeling, and AI-powered decision-making to optimize credit risk management strategies and achieve business objectives.

## 6 Conclusion

Integrating Big Data into credit risk assessment is revolutionizing financial institutions' operations. Our study reviewed 50 papers from 2019 to 2025, highlighting how the 5Vs of Big Data (Volume, Variety, Velocity, Veracity, and Value) enhance predictive accuracy and financial inclusivity. By leveraging extensive datasets, including billions of transactions (Volume) and diverse sources like social media and utility payments (Variety), institutions achieve more precise and timely risk profiling. Real-time data processing (Velocity) allows immediate adjustments to risk profiles based on current financial activities. Rigorous validation techniques ensure data accuracy (Veracity), while advanced analytics and machine learning extract actionable insights (Value), leading to better decisions. Despite challenges related to data quality, privacy, and technological investment, Big Data's benefits are significant. Future efforts should prioritize overcoming these obstacles, ensuring data accuracy and privacy, and enhancing integration methods. Embracing Big Data is crucial for financial institutions to stay competitive, meet diverse borrower needs, and foster a more inclusive and stable financial landscape.

## References

- [1] Burra RS, APCV GR, Vellela SS. Strategic Insights: Unleashing the Power of Big Data Analytics for Credit Investigation and Risk Mitigation in Commercial Banking. *International Journal of Progressive Research in Engineering Management and Science*. 2024;4(01):458–464.
- [2] Du G, Liu Z, Lu H. Application of innovative risk early warning mode under big data technology in Internet credit financial risk assessment. *Journal of Computational and Applied Mathematics*. 2021;386:113260. <https://doi.org/https://doi.org/10.1016/j.cam.2020.113260>.
- [3] Gu Z, Lv J, Wu B, Hu Z, Yu X. Credit risk assessment of small and micro enterprise based on machine learning. *Heliyon*. 2024;10(5):e27096.

- [4] Guo Y. Credit risk assessment of P2P lending platform towards big data based on BP neural network. *Journal of Visual Communication and Image Representation*. 2020;71:102730. <https://doi.org/https://doi.org/10.1016/j.jvcir.2019.102730>.
- [5] Hindistan YS, Kiyakoğlu BY, Rezaeinazhad AM, Korkmaz HE, Dağ H. Alternative Credit Scoring and Classification Employing Machine Learning Techniques on a Big Data Platform. In: 2019 4th International Conference on Computer Science and Engineering (UBMK); 2019. p. 1–4.
- [6] Lin M, Chen J. Research on Credit Big Data Algorithm Based on Logistic Regression. *Procedia Computer Science*. 2023;228:511–518.
- [7] Mingmin G, Liyuan C, Yihan L, Xin H, Jian W, Yuchen L. Based on the Development of “Blockchain and Big Data” Personal Credit System Research. In: 2022 International Symposium on Electrical, Electronics and Information Engineering (ISEEIE); 2022. p. 19–24.
- [8] Niu A, Cai B, Cai S. [Retracted] Big Data Analytics for Complex Credit Risk Assessment of Network Lending Based on SMOTE Algorithm. *Complexity*. 2020;2020(1):8563030.
- [9] Óskarsdóttir M, Bravo C, Sarraute C, Vanthienen J, Baesens B. The value of big data for credit scoring: Enhancing financial inclusion using mobile phone data and social network analytics. *Applied Soft Computing*. 2019;74:26–39.
- [10] Ren J. Research on Enterprise Credit Risk Assessment Based on Big Data of Electricity. In: 2023 IEEE 11th Joint International Information Technology and Artificial Intelligence Conference (ITAIC). vol. 11; 2023. p. 53–57.
- [11] Wang F, Ding L, Yu H, Zhao Y. Big data analytics on enterprise credit risk evaluation of e-Business platform. *Information Systems and e-Business Management*. 2020;18(3):311–350.
- [12] Huang Y, Li Z, Qiu H, Tao S, Wang X, Zhang L. BigTech credit risk assessment for SMEs. *China Economic Review*. 2023;81:102016.
- [13] Nahar J, Rahaman MA, Alauddin M, Rozony FZ. Big Data in Credit Risk Management: A Systematic Review Of Transformative Practices And Future Directions. *International Journal of Management Information Systems and Data Science*. 2024;1(04):68–79.
- [14] Patel K. Credit card analytics: a review of fraud detection and risk assessment techniques. *International Journal of Computer Trends and Technology*. 2023;71(10):69–79.
- [15] Addy WA, Ugochukwu CE, Oyewole AT, Ofodile OC, Adeoye OB, Okoye CC. Predictive analytics in credit risk management for banks: A comprehensive review. *GSC Advanced Research and Reviews*. 2024;18(2):434–449.
- [16] Zhang Y, Ye S, Liu J, Du L. Impact of the development of FinTech by commercial banks on bank credit risk. *Finance Research Letters*. 2023;55:103857.
- [17] Xu R, Zhao D. Private Equity in the Age of Fintech. In: *Digital Transformation of Private Equity in China*. Springer; 2023. p. 1–13.
- [18] Aziz Z, Bestak R. Insight into Anomaly Detection and Prediction and Mobile Network Security Enhancement Leveraging K-Means Clustering on Call Detail Records. *Sensors*. 2024;24(6):1716.
- [19] Duanmu J, McBrayer GA. Structural Drivers of Credit Rating Uncertainty: An Examination of the Changes Imposed by Dodd-Frank. *Journal of Financial Services Research*. 2024;65(2):243–267.
- [20] Santosh T, Ramesh D. Machine learning approach on apache spark for credit card fraud detection. *Ingenierie des Systemes d’Information*. 2020;25:101–106.

- [21] Kurt S, Heitz J, Bundi N, Breymann W. Large-scale data-driven financial risk modeling using big data technology. In: 2018 IEEE/ACM 5th International Conference on Big Data Computing Applications and Technologies (BDCAT). IEEE; 2018. p. 206–207.
- [22] Mohamed A, Najafabadi MK, Wah YB, Zaman EAK, Maskat R. The state of the art and taxonomy of big data analytics: view from new big data framework. *Artificial intelligence review*. 2020;53:989–1037.
- [23] Hindistan YS, Kiyakoğlu BY, Rezaeinazhad AM, Korkmaz HE, Dağ H. Alternative credit scoring and classification employing machine learning techniques on a big data platform. In: 2019 4th International Conference on Computer Science and Engineering (UBMK). IEEE; 2019. p. 1–4.
- [24] Seo YD, Ahn JH. Hadoop-based integrated monitoring platform for risk prediction using big data. *Applied Mechanics and Materials*. 2016;826:113–117.
- [25] Mostafaeipour A, Jahangard Rafsanjani A, Ahmadi M, Arockia Dhanraj J. Investigating the performance of Hadoop and Spark platforms on machine learning algorithms. *The Journal of Supercomputing*. 2021;77:1273–1300.
- [26] Wang L. Application of Machine Learning in Risk Assessment of Big Data IOT Credit Financial Management of Operator. In: 2022 2nd International Conference on Networking Systems of AI (INSAI); 2022. p. 228–232.
- [27] Zhang B. Application of Innovative Risk Early Warning Model Based on Big Data Technology in Internet Credit Financial Risk. *Journal of Information Technology Research (JITR)*. 2022;15(1):1–12.
- [28] Zhou X. Research on Personal Credit Risk Assessment Index System Based on Internet Social Big Data. In: 2019 IEEE 4th Advanced Information Technology, Electronic and Automation Control Conference (IAEAC); 2019. p. 2704–2708.
- [29] Wu M, Bu Y. Research on Credit Resources Management of China's Commercial Banks Based on the Application of Big Data Credit Investigation. In: 2021 IEEE 6th International Conference on Cloud Computing and Big Data Analytics (ICCCBDA); 2021. p. 27–31.
- [30] Xu C, Tao J, Xu P. Credit Decision System of Small and Medium Sized Micro Enterprises Based on Big Data Technology and Risk Assessment Thinking. In: 2021 6th International Conference on Smart Grid and Electrical Automation (ICSGEA); 2021. p. 477–480.
- [31] Xianyu Q, Hai M. Research on Default Prediction Model of Corporate Credit Risk Based on Big Data Analysis Algorithm. *Procedia Computer Science*. 2023;221:300–307.
- [32] Gao L, Xiao J. Big data credit report in credit risk management of consumer finance. *Wireless Communications and Mobile Computing*. 2021;2021(1):4811086.
- [33] Sun M, Sun W, Sun Y, Liu S, Jiang M, Xu Z. Applying Hybrid Graph Neural Networks to Strengthen Credit Risk Analysis. *arXiv preprint arXiv:241004283*. 2024;.
- [34] Wu W. Credit risk measurement, decision analysis, transformation and upgrading for financial big data. *Complexity*. 2022;2022(1):8942773.
- [35] Wu Y, Xi Z, Wang W, Li Q. Enterprise Credit Rating Model Based on Long and Short-Term Trend of Desensitized Power Load Data. In: *Proceedings of the 2023 3rd International Conference on Big Data, Artificial Intelligence and Risk Management. ICBAR '23*. New York, NY, USA: Association for Computing Machinery; 2024. p. 1060–1064. Available from: <https://doi.org/10.1145/3656766.3656941>.
- [36] Li S. Leveraging Big Data for SME Credit Risk Assessment: A Novel BP-KMV and GARCH Integration. *Journal of the Knowledge Economy*. 2024;p. 1–29.

- [37] Yang F, Xu J. Optimizing bank credit risk assessment models using big data analytics. *Applied Mathematics and Nonlinear Sciences*. 2024;9(1).
- [38] Song L, Li H, Tan Y, Li Z, Shang X. Enhancing enterprise credit risk assessment with cascaded multi-level graph representation learning. *Neural Networks*. 2024;169:475–484.
- [39] Emmanuel I, Sun Y, Wang Z. A machine learning-based credit risk prediction engine system using a stacked classifier and a filter-based feature selection method. *Journal of Big Data*. 2024;11(1):23.
- [40] Ramesh R, Jeyakarthic M. Enhancing credit risk prediction with hybrid deep learning and sand cat swarm feature selection. *Multimedia Tools and Applications*. 2024;83(21):60243–60263.
- [41] Zhu X, Qin L, Samuri SM, Adnan MHM. Automotive Consumer Loans Risk Assessment Predictive Modeling Using Generative Adversarial Network and Stacked Autoencoder Neural Networks. *Journal of Advanced Research in Applied Sciences and Engineering Technology*. 2025;51(2):45–56.
- [42] Yang X. Financial Risk Assessment Model Based on Fuzzy Logic. *Journal of Electrical Systems*. 2024;20(1).
- [43] Thirumagal PG, Vaddepalli S, Das T, Das S, Madem S, Immaculate PS. AI-Enhanced IoT Data Analytics for Risk Management in Banking Operations. In: 2024 5th International Conference on Recent Trends in Computer Science and Technology (ICRTCST); 2024. p. 177–181.
- [44] Hassan MAM, T RG, Mansur UM, Jha R, Fahim MFH, R MT. Interpretable Machine Learning Models for Credit Risk Assessment. In: 2024 11th International Conference on Computing for Sustainable Global Development (INDIACom); 2024. p. 361–365.
- [45] Mitra R, Dongre A, Dangare P, Goswami A, Tiwari MK. Knowledge graph driven credit risk assessment for micro, small and medium-sized enterprises. *International Journal of Production Research*. 2024;62(12):4273–4289.
- [46] Huang B, Zhao F, Tian M, Zhang D, Zhang X, Wang Z, et al. Explainability of Machine Learning in Credit Risk Assessment of SMEs. In: *Artificial Intelligence and Human-Computer Interaction*. IOS Press; 2024. p. 165–176.
- [47] Palaiokrassas G, Scherrers S, Makri E, Tassiulas L. Machine Learning in DeFi: Credit Risk Assessment and Liquidation Prediction. In: 2024 IEEE International Conference on Blockchain and Cryptocurrency (ICBC); 2024. p. 650–654.
- [48] Wang L, Zheng J, Yao J, Chen Y. A multi-stage integrated model based on deep neural network for credit risk assessment with unbalanced data. *Kybernetes*. 2024;.
- [49] Kotti J, Ganesh CN, Naveenan R, Gorde SG, Basha M, Pramanik S, et al. Utilizing Big Data Technology for Online Financial Risk Management. In: *Artificial Intelligence Approaches to Sustainable Accounting*. IGI Global; 2024. p. 135–148.
- [50] Peng Y. Construction and Evaluation of Credit Risk Early Warning Indicator System of Internet Financial Enterprises Based On AI and Knowledge Graph Theory. *Procedia Computer Science*. 2024;243:918–927.
- [51] Talaat FM, Aljadani A, Badawy M, Elhosseini M. Toward interpretable credit scoring: integrating explainable artificial intelligence with deep learning for credit card default prediction. *Neural Computing and Applications*. 2024;36(9):4847–4865.
- [52] Xu T. Credit Risk Assessment Using a Combined Approach of Supervised and Unsupervised Learning. *Journal of Computational Methods in Engineering Applications*. 2024;p. 1–12.

- [53] Hou L, Bi G, Guo Q. An improved sparrow search algorithm optimized LightGBM approach for credit risk prediction of SMEs in supply chain finance. *Journal of Computational and Applied Mathematics*. 2025;454:116197.
- [54] Kwon S, Jang J, Kim CO. Credit scoring using multi-task Siamese neural network for improving prediction performance and stability. *Expert Systems with Applications*. 2025;259:125327.
- [55] Chang V, Xu QA, Akinloye SH, Benson V, Hall K. Prediction of bank credit worthiness through credit risk analysis: an explainable machine learning study. *Annals of Operations Research*. 2024;p. 1–25.
- [56] Yu C, Jin Y, Xing Q, Zhang Y, Guo S, Meng S. Advanced user credit risk prediction model using lightgbm, xgboost and tabnet with smoteenn. *arXiv preprint arXiv:240803497*. 2024;.
- [57] Al Shiam SA, Hasan MM, Pantho MJ, Shochona SA, Nayeem MB, Choudhury MTH, et al. Credit Risk Prediction Using Explainable AI. *Journal of Business and Management Studies*. 2024;6(2):61–66.
- [58] Yang F, Abedin MZ, Hajek P. An explainable federated learning and blockchain-based secure credit modeling method. *European Journal of Operational Research*. 2024;317(2):449–467.
- [59] du Toit HA, Schutte WD, Raubenheimer H. Integrating traditional and non-traditional model risk frameworks in credit scoring. *South African Journal of Economic and Management Sciences*. 2024;27(1):13.
- [60] Bussmann N, Giudici P, Marinelli D, Papenbrock J. Explainable machine learning in credit risk management. *Computational Economics*. 2021;57(1):203–216.
- [61] Babaei G, Giudici P, Raffinetti E. Explainable fintech lending. *Journal of Economics and Business*. 2023;125:106126.
- [62] Chen Y, Giudici P, Liu K, Raffinetti E. Measuring fairness in credit ratings. *Expert Systems with Applications*. 2024;258:125184.
- [63] Chai N, Abedin MZ, Yang L, Shi B. Farmers’ credit risk evaluation with an explainable hybrid ensemble approach: A closer look in microfinance. *Pacific-Basin Finance Journal*. 2025;89:102612.
- [64] Lin SC, Tian F, Wang K, Zhao X, Huang J, Xie Q, et al.: Open FinLLM Leaderboard: Towards Financial AI Readiness. Available from: <https://arxiv.org/abs/2501.10963>.